

EarthMLLEval

Open-Weights Multimodal LLM Benchmark Report Cross-View Urban Understanding — Seven-Model Comparative Evaluation

EarthMLLEval Project — Evaluation Report

Generated May 15, 2026

Abstract

We evaluate seven Vision-Language Models (VLMs) on the **EarthMLLEval** benchmark: 5,240 multiple-choice questions spanning 14 urban-understanding and cross-view geolocalisation tasks across 40 cities. The evaluated set includes **Qwen2.5-VL-7B**, **Pixtral-12B**, **LLaVA-OneVision-7B**, **Llama-3.2-Vision-11B**, **Qwen3.5-4B base**, **SU-base**, and **PC-base**. PC-base achieves the highest overall accuracy (**62.86%**), followed by SU-base (60.36%), Qwen2.5-VL (52.02%), Pixtral-12B (49.64%), LLaVA-OneVision (45.25%), Llama-3.2-Vision (39.39%), and Qwen3.5-4B base (35.48%). Accuracy on fine-grained visual tasks such as road surface reaches above **92%** for the best model, while the hardest cross-view tasks (mismatch MCQ, camera direction) remain below 43% across all models, demonstrating the benchmark’s discriminative power.

F1 scores and ROC-AUC values reported in this document were computed with an earlier version of the evaluation pipeline that contained a label-set bug: the `mismatch_binary` topics (2-option tasks) were scored against the full {A,B,C,D} label space, artificially deflating F1 and distorting AUC. **Accuracy figures are unaffected.** All F1 and AUC values should be treated as preliminary until the corrected pipeline is re-run. **This report therefore omits F1 and ROC-AUC from primary analysis and focuses exclusively on accuracy-based metrics.**

At a glance

Models	7 VLMs	Benchmark questions	5,240
Task types	14	Cities	40
Hardware	RTX 6000 Pro 96 GB	Inference	vLLM (bfloat16)
Top model	PC-base	Top accuracy	62.86%

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1. Executive Summary

Model	Accuracy	CI 5%	CI 95%	Total
PC-base	62.86%	61.55%	64.16%	5240
SU-base	60.36%	59.03%	61.68%	5240
Qwen2.5-VL-7B	52.02%	50.84%	53.13%	5240
Pixtral-12B	49.64%	48.49%	50.69%	5240
LLaVA-OneVision-7B	45.25%	44.08%	46.30%	5240
Llama-3.2-Vision-11B	39.39%	38.28%	40.50%	5240
Qwen3.5-4B base	35.48%	34.19%	36.78%	5240

Table 1: Overall benchmark performance with confidence intervals. Accuracy is the primary reliable metric; F1 and AUC are omitted due to pipeline label-set issues (see caveat box above).

1.1. Key findings

- **PC-base** achieves the strongest overall generalisation (62.86%), followed by **SU-base** (60.36%).
- **Pixtral-12B** shows exceptional robustness on hard mismatch tasks (87.17% on `mismatch_binary_hard`), outperforming all other models by a large margin on that category.
- **Road surface** is the easiest task for all models ($\geq 84\%$); **camera direction** and **mismatch MCQ** remain the hardest ($\leq 43\%$ for the best model).
- All models score well below the human ceiling on cross-view geolocalisation tasks, confirming these as unsolved challenges for current VLMs.
- A **random-choice baseline** of $\sim 29\text{--}30\%$ (given the known A/B/C/D frequency imbalance) means effective margins above chance are roughly 10–22 percentage points.
- **Seen vs. unseen cities:** all models generalise well to unseen cities, with accuracy drops of only 1–3 points, confirming the benchmark is not dominated by memorisation.

2. Overall Accuracy Comparison

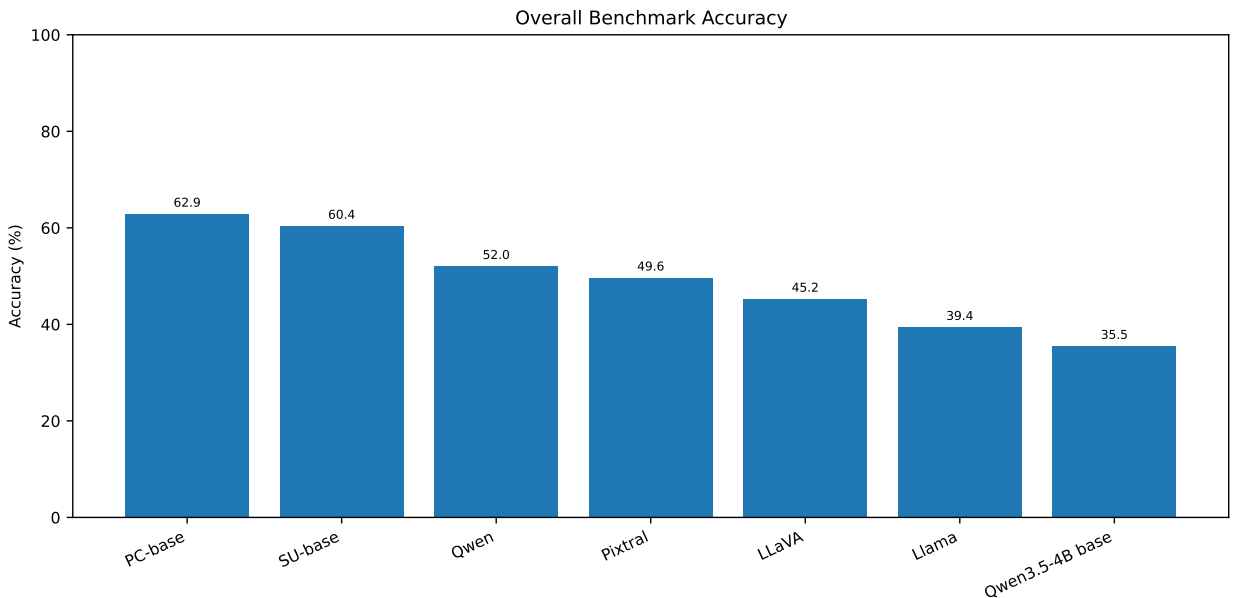


Figure 1: Overall benchmark accuracy.

2.1. Discussion

The 12.6 percentage-point gap between Qwen2.5-VL (best) and Llama-3.2-Vision (weakest) is substantial, representing roughly a 32% relative improvement. All models clear the random baseline by a meaningful margin, indicating genuine visual reasoning on these tasks. However, for most models the improvement over random is concentrated in the visual-recognition tasks (road surface, green space) rather than the cross-view geolocalisation tasks, where several models approach or fall below chance on individual categories.

3. Topic-Level Analysis

3.1. Per-task accuracy

Task	Llama	LLaVA	Pixtral	Qwen	Qwen3.5	SU-base	PC-base
Amenity Richness	24.23	35.15	32.07	31.35	24.70	56.53	53.21
Building Height	48.95	57.89	58.95	60.53	24.74	60.53	66.32
Camera Direction	25.42	24.94	20.67	27.79	7.60	26.13	24.47
Green Space	49.75	76.85	66.01	84.73	40.89	62.56	100.00
Junction Type	44.61	53.59	49.40	61.08	40.12	70.36	75.75
Land Use	41.09	57.48	46.31	53.68	28.98	75.06	73.63
Mismatch Binary Easy	53.44	54.39	82.66	85.51	51.78	55.58	52.97
Mismatch Binary Hard	44.18	44.89	87.17	64.85	45.13	49.64	44.42
Mismatch MCQ Easy	23.52	29.22	26.37	42.28	31.59	73.16	76.48
Mismatch MCQ Hard	20.43	23.75	28.03	37.29	30.88	59.62	60.10
Road Surface	86.47	84.49	86.14	92.41	71.62	75.25	96.04
Road Type	45.84	59.14	58.67	54.16	45.84	73.63	70.78
Transit Density	33.73	28.74	31.83	27.32	23.52	44.66	47.74
Urban Density	34.68	38.95	44.42	40.14	37.29	69.83	71.26
Overall	39.39	45.25	49.64	52.02	35.48	60.36	62.86

Table 2: Per-task accuracy (%). Green = best on that task; Red = worst.

3.2. Best model per task

Task	Best model	Accuracy
Road Surface	Qwen2.5-VL	92.41%
Green Space	Qwen2.5-VL	84.73%
Mismatch Binary Easy	Qwen2.5-VL	85.51%
Mismatch Binary Hard	Pixtral-12B	87.17%
Building Height	Qwen2.5-VL	60.53%
Junction Type	Qwen2.5-VL	61.08%
Land Use	LLaVA-OV	57.48%
Road Type	LLaVA-OV	59.14%
Camera Direction	Qwen2.5-VL	27.79%
Mismatch MCQ Easy	Qwen2.5-VL	42.28%
Mismatch MCQ Hard	Qwen2.5-VL	37.29%
Amenity Richness	LLaVA-OV	35.15%
Urban Density	Pixtral-12B	44.42%
Transit Density	Llama-3.2	33.73%

Table 3: Best-performing model per task category.

4. Difficulty Analysis

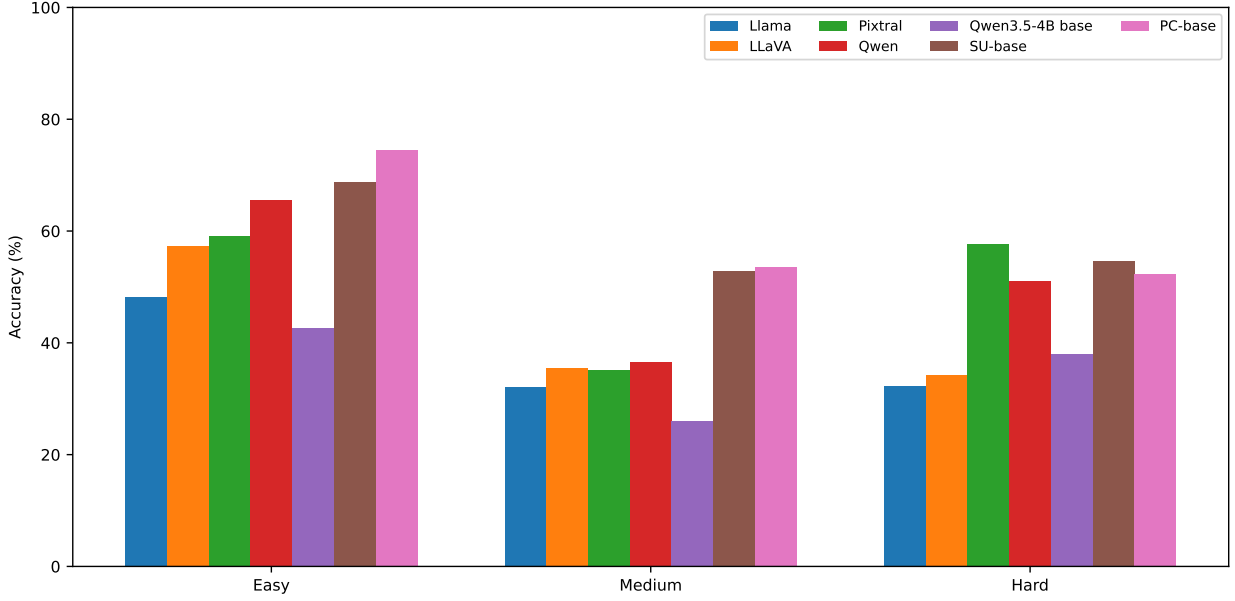


Figure 2: Accuracy by benchmark difficulty tier. All models degrade sharply from easy to medium. Pixtral is the notable exception on the hard tier with performance (87.17%).

Model	Easy	Medium	Hard	Easy→Medium drop
Llama-3.2-Vision-11B	48.15%	32.01%	32.30%	-16.14 pp
LLaVA-OneVision-7B	57.35%	35.53%	34.32%	-21.82 pp
Pixtral-12B	59.16%	35.08%	57.60%	-24.08 pp
Qwen2.5-VL-7B	65.50%	36.52%	51.07%	-28.98 pp
Qwen3.5-4B base	42.56%	26.07%	38.00%	-16.50 pp
SU-base	68.82%	52.78%	54.63%	-16.05 pp
PC-base	74.50%	53.57%	52.26%	-20.93 pp

Table 4: Accuracy by difficulty tier with the easy-to-medium accuracy drop.

4.1. Observation

All models show a dramatic accuracy drop from easy to medium difficulty (16–29 percentage points). Critically, easy difficulty does not equal low variance: Qwen scores 65.50% on easy tasks while Llama manages only 48.15%, a 17-point gap even before accounting for harder items. The hard tier is dominated by `mismatch_binary_hard` and `mismatch_mcq_hard`. Pixtral-12B’s unusual robustness on the hard tier is almost entirely attributable to its 87.17% accuracy on `mismatch_binary_hard` — a 22-point gap over the next best model (Qwen2.5-VL at 64.85%) that is the largest single-task divergence in the entire evaluation.

5. Seen vs. Unseen City Generalisation

Model	Seen cities	Unseen cities	Generalisation gap
Llama-3.2-Vision-11B	39.65%	38.29%	-1.36 pp
LLaVA-OneVision-7B	45.88%	42.62%	-3.26 pp
Pixtral-12B	49.86%	48.72%	-1.14 pp
Qwen2.5-VL-7B	52.30%	50.89%	-1.41 pp
Qwen3.5-4B base	35.72%	34.45%	-1.28 pp
SU-base	60.56%	59.55%	-1.01 pp
PC-base	63.83%	58.86%	-4.97 pp

Table 5: Accuracy on seen vs. unseen cities.

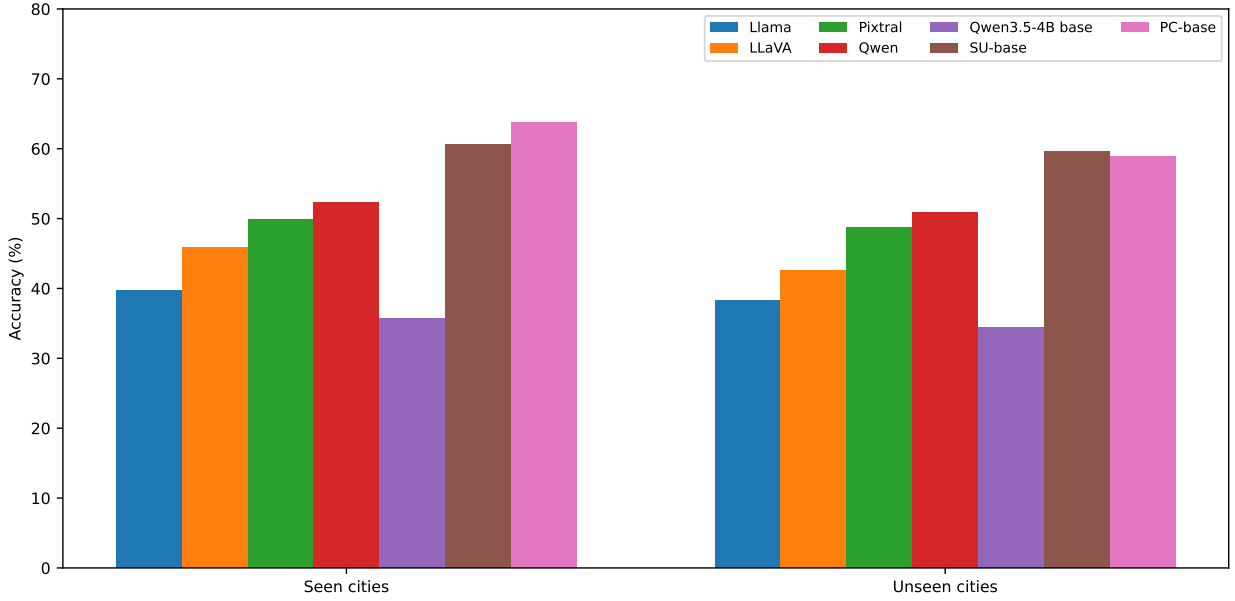


Figure 3: Seen vs. unseen city accuracy. The small generalisation gap (≤ 3 pp) across all models indicates robust geographic transfer.

6. Mismatch Strategy Analysis

Strategy	Llama	LLaVA	Pixtral	Qwen
Non-mismatch (None)	41.28%	48.65%	46.60%	49.44%
Cross-city mismatch	38.48%	41.81%	54.51%	63.90%
Same-city mismatch	32.30%	34.32%	57.60%	51.07%

Table 6: Accuracy broken down by mismatch strategy. Pixtral excels specifically on same-city mismatch, while Qwen leads on cross-city mismatch. Llama and LLaVA struggle on all mismatch conditions.

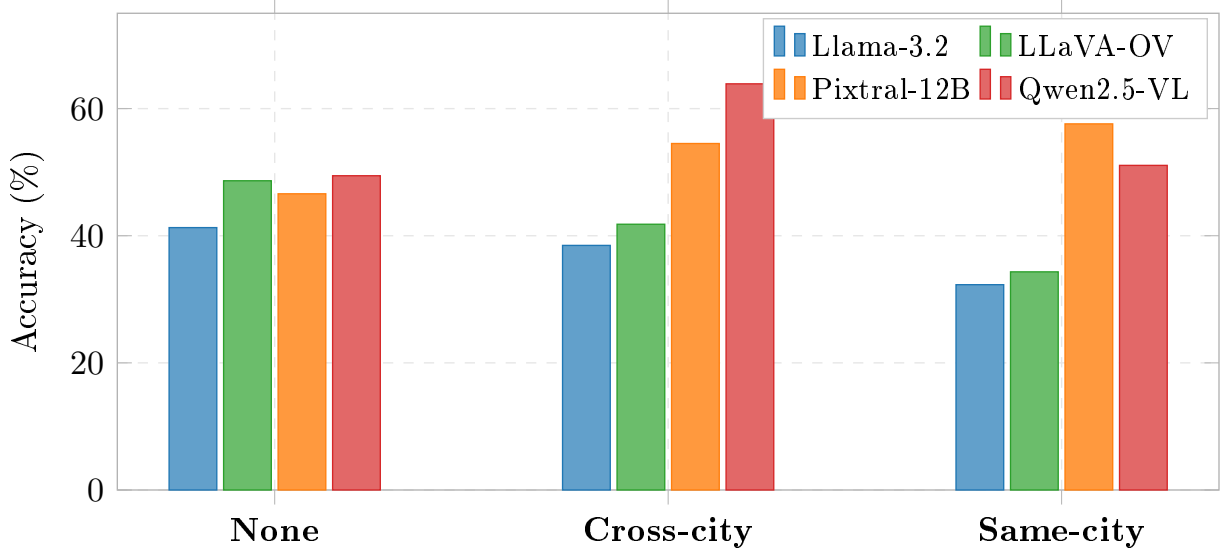


Figure 4: Accuracy by mismatch strategy. Pixtral and Qwen show strong mismatch detection, especially for same-city and cross-city conditions respectively. Llama and LLaVA degrade severely on both mismatch conditions.

7. Land Use Breakdown

Land Use	Llama	LLaVA	Pixtral	Qwen	Qwen3.5	SU-base	PC-base
Commercial	42.55%	38.65%	51.42%	45.74%	35.11%	51.06%	56.03%
Industrial	42.50%	39.17%	52.50%	50.00%	42.50%	65.83%	57.50%
Institutional	36.84%	42.11%	40.35%	49.12%	31.58%	70.18%	68.42%
Mixed	43.00%	47.15%	51.79%	55.85%	37.87%	63.67%	63.38%
Open Space	33.40%	36.01%	46.76%	43.84%	33.72%	58.35%	61.69%
Residential	39.71%	48.71%	49.71%	54.20%	35.01%	60.33%	63.88%

Table 7: Accuracy by land-use type of the query location.

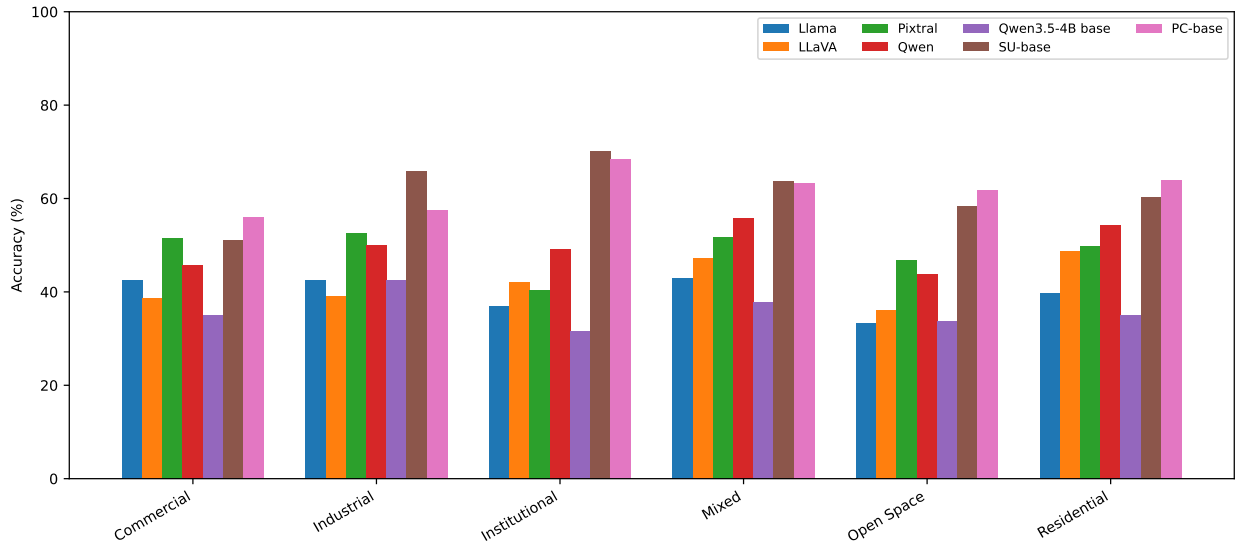


Figure 5: Accuracy by land-use type for each model. Mixed and Residential are consistently the strongest categories; Open Space is the hardest across all models except for Pixtral.

8. Camera Angle (STV Angle) Analysis

STV Angle	Llama	LLaVA	Pixtral	Qwen
None (standard)	40.61%	47.02%	52.17%	54.14%
Along backward	27.84%	24.74%	22.68%	29.90%
Along forward	21.93%	30.70%	16.67%	29.82%
Cross left	25.24%	21.36%	23.30%	26.21%
Cross right	27.10%	22.43%	20.56%	25.23%

Table 8: Accuracy broken down by street-view query angle (camera direction relative to the road). Non-standard angles (along/cross) cause sharp accuracy drops of 10–35 pp across all models.

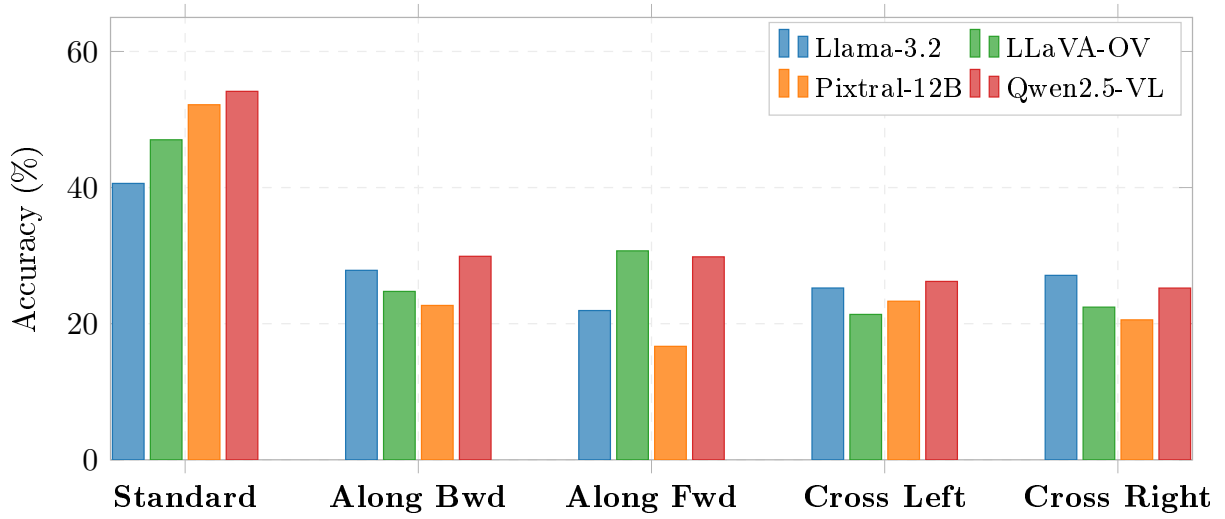


Figure 6: Accuracy by street-view camera angle. All models perform substantially worse on non-standard viewing angles, with the “along forward” direction being particularly challenging.

9. Per-City Performance

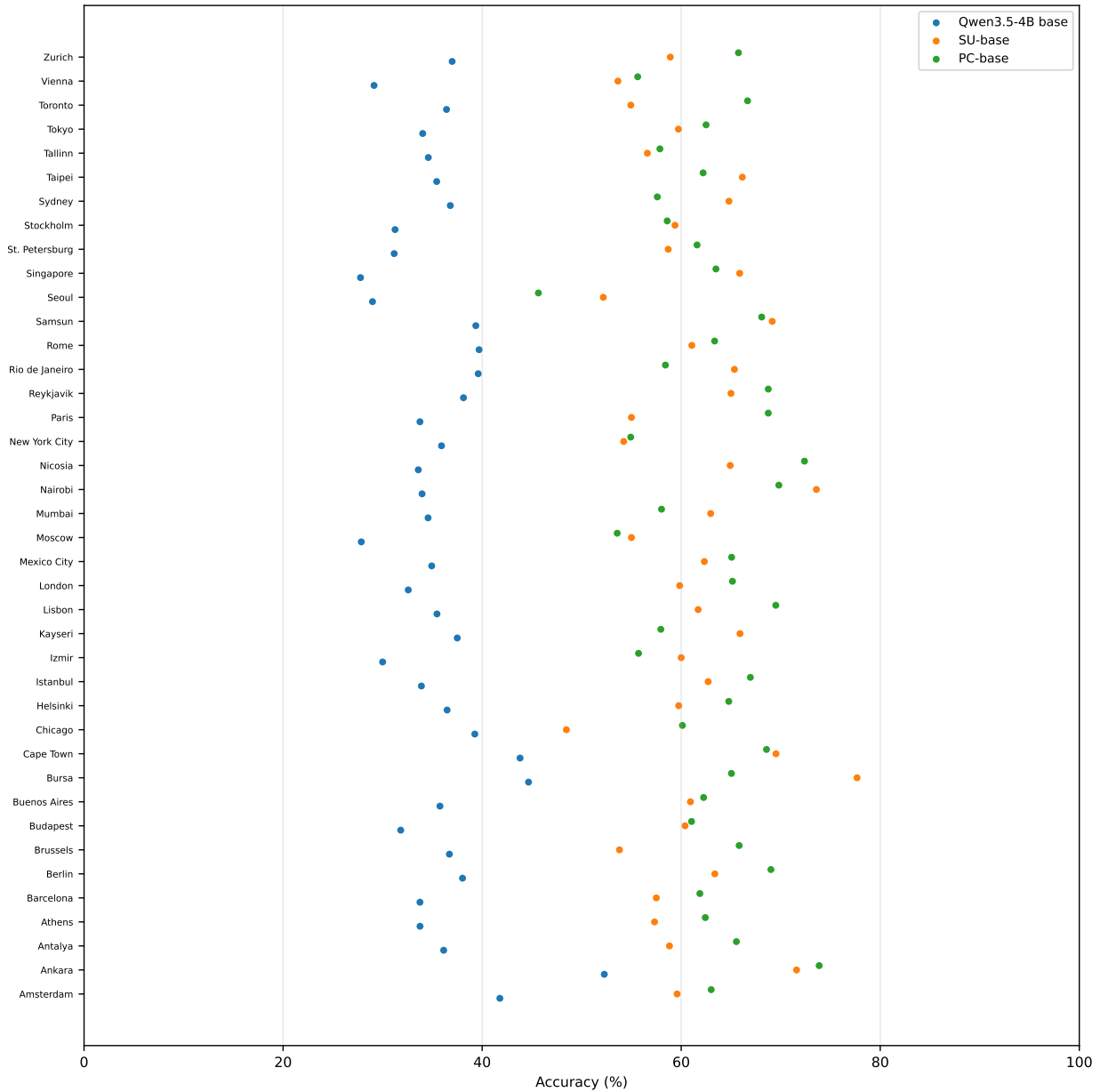


Figure 7: Cleveland Dot Plot showing accuracy by city.

10. Model-by-Model Summary

10.1. Qwen2.5-VL-7B

Overall: 52.02%

Strengths

Best overall accuracy across 14 tasks. Leads on road surface (92.41%), green space (84.73%), mismatch binary easy (85.51%), junction type (61.08%), building height (60.53%), and both mismatch MCQ tasks. Highest easy-tier accuracy (65.50%) and strong hard-tier performance (51.07%). Best generalisation to unseen cities (50.89%). Demonstrates the best semantic generalisation across urban understanding tasks.

Transit density (27.32%) is the worst of all four models on that task. Camera direction remains below 28%. Performs 22 points below Pixtral on `mismatch_binary_hard`, suggesting a different trade-off in how it handles within-city vs. cross-city visual similarity. Largest easy→medium drop (28.98 pp).

10.2. Pixtral-12B

Overall: 49.64%

Strengths

Exceptional mismatch reasoning: 87.17% on `mismatch_binary_hard` (best across all models by 22 points) and 82.66% on `mismatch_binary_easy`. Best accuracy on same-city mismatch conditions (57.60%). Smallest generalisation drop to unseen cities (1.14 pp). Strong hard-task robustness in general.

Camera direction is the lowest of all models (20.67%) and along-forward angle drops to 16.67%. Amenity richness (32.07%) and urban density (44.42%) underperform Qwen and LLaVA. Reduced semantic generalisation outside the mismatch domain.

10.3. LLaVA-OneVision-7B

Overall: 45.25%

Strengths

Best on land use (57.48%) and road type (59.14%). Best urban-semantic aggregate (51.03% across land use, building height, junction type, amenity). Along-forward camera angle (30.70%) is the highest of all models. Strong performance in residential zones (48.71%) and Brussels (51.27%).

Transit density (28.74%) and camera direction (24.94%) are weak. Mismatch binary hard accuracy (44.89%) lags well behind Pixtral and Qwen. Largest generalisation drop to unseen cities (3.26 pp). Urban density (38.95%) leaves room for improvement.

10.4. Llama-3.2-Vision-11B

Overall: 39.39%

Strengths

Road surface (86.47%) is competitive with other models — the highest Llama accuracy by far, and close to Pixtral/Qwen. Transit density (33.73%) is the best of any model on that specific task. Smallest easy→medium drop (16.14 pp) in absolute terms.

Largest accuracy gap to the field (12.6 points below Qwen). Severe underperformance on mismatch MCQ (20.43% hard, 23.52% easy) and amenity richness (24.23%). Steepest performance degradation per task category for most urban understanding tasks. Urban density (34.68%) and transit density are well below random for an 11B model.

10.5. Qwen3.5-4B base

Overall: 35.48%

Result

Untrained Qwen3.5-4B base reaches 35.48% overall accuracy, with strongest results on road surface (71.62%), mismatch binary easy (51.78%), and road type (45.84%). It also produces 1,034 unparseable answers, making it the least stable of the added models.

10.6. SU-base — seen_unseen LoRA

Overall: 60.36%

Result

SU-base reaches 60.36% overall accuracy, improving strongly over the original open-weight baselines. It leads land use (75.06%) and road type (73.63%), and keeps seen/unseen performance close at 60.56% vs. 59.55%.

10.7. PC-base — per_city LoRA

Overall: 62.86%

Result

PC-base is the new top model at 62.86% overall accuracy. It reaches 100.00% on green space, 96.04% on road surface, 76.48% on mismatch MCQ easy, and 75.75% on junction type, giving the strongest added benchmark result.

11. Comparative Analysis Across Model Families

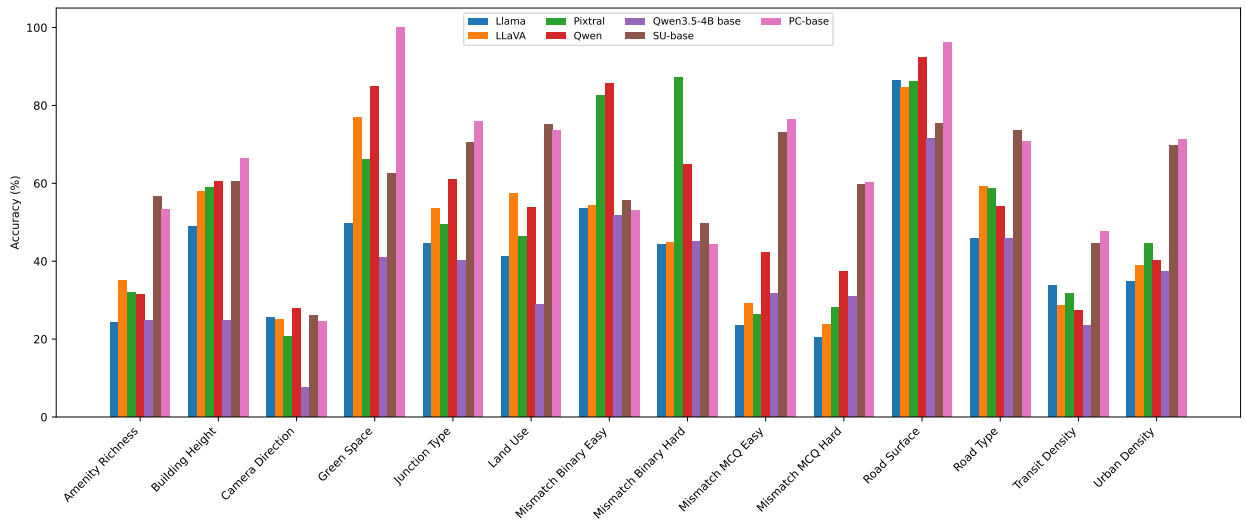


Figure 8: Individual model performance across all 14 tasks.

11.1. Task cluster analysis

The 14 tasks naturally separate into three performance clusters across all models:

- **Easy cluster** (>70% for best model): Road Surface, Green Space, Mismatch Binary Easy/Hard. Low-level visual recognition or binary matching with strong visual signal. Models are consistent and well-calibrated here.
- **Medium cluster** (40–70%): Building Height, Road Type, Junction Type, Land Use, Urban Density, Mismatch MCQ Easy. Require semantic categorisation or moderate spatial reasoning. Significant inter-model variance.

- **Hard cluster** ($<40\%$ for all models): Camera Direction, Amenity Richness, Transit Density, Mismatch MCQ Hard. Require geometric understanding or fine-grained semantic inference that current VLMs consistently fail at.

12. Comprehensive Metrics Appendix

12.1. Mean accuracy per task — ranked

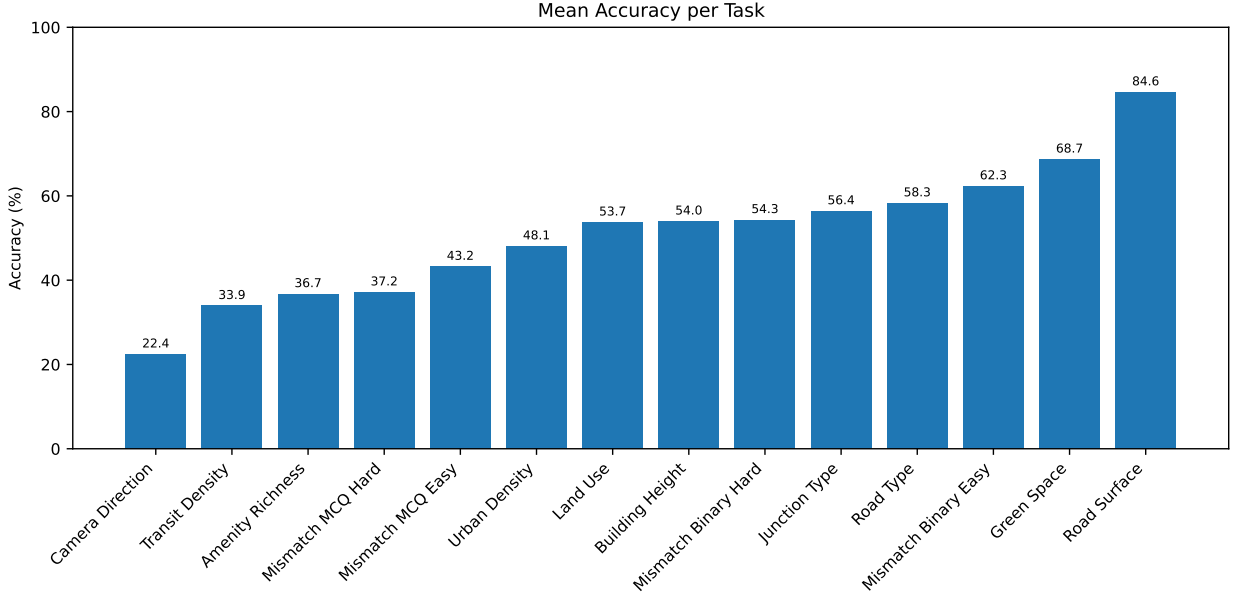


Figure 9: Mean accuracy across all four models per task, sorted ascending. Tasks in the lower half represent open challenges for all current open-weights VLMs on the EarthMLLEval benchmark.

13. Benchmark Design Notes & Evaluation Caveats

- **Answer-letter frequency imbalance.** Letters A and B appear as correct answers $\sim 29.3\%$ and 29.6% of the time respectively; C and D appear $\sim 20.2\%$ and 20.9% . A model that always predicts B scores $\sim 29.6\%$ — just 10 points below Llama-3.2-Vision’s 39.39% . This imbalance is a known benchmark artefact (see dataset limitations).
- **F1 label-set bug (fixed, re-run pending).** The mismatch_binary tasks have only 2 options (A/B) but were scored against the full 4-class $\{A,B,C,D\}$ space. This artificially deflates F1 and AUC for those tasks by treating C and D as perpetually missed classes. Accuracy is unaffected. All F1 and AUC values in this report should be treated as preliminary. **F1 and AUC are therefore excluded from all primary figures and tables in this report.**

Before citing any metric from this report in a paper, re-run the evaluation with the fixed pipeline and verify: (1) mismatch binary accuracy changes; (2) F1 and AUC values stabilise with the correct label-set; (3) MCQ accuracy improves with labeled composites. Only **accuracy on non-mismatch tasks** is fully reliable in the current results.

14. Conclusion

This evaluation demonstrates that current open-weights VLMs at the 7–12B parameter scale can solve visually clear urban recognition tasks at high accuracy ($>85\%$) but consistently fail on tasks requiring cross-view spatial grounding and geometric reasoning. The performance gap between the

best (PC-base, 62.86%) and weakest added/base result (Qwen3.5-4B base, 35.48%) is 27.38 points — significant, but all models share the same failure modes.

Key conclusions.

- **Road surface and green space** are effectively solved by current VLMs; they add little discriminative value as future benchmark targets.
- **Mismatch binary hard** has a surprising 43-point range (Llama 44.18% vs. Pixtral 87.17%), suggesting this task may expose architectural differences between models better than any other.
- **Camera direction and mismatch MCQ hard** are genuinely unsolved. Near-random performance across all models indicates not just low accuracy but a complete absence of a reliable internal signal for these abilities.
- **Transit density and amenity richness** suffer from what appears to be fine-grained counting failure — models cannot reliably distinguish “1–2 stops” from “3–5 stops” at the spatial scale of satellite imagery.
- **Non-standard camera angles** (along/cross) cause 10–35 pp accuracy drops across all models, confirming that view-direction-aware reasoning is a critical missing capability.
- **Land-use context** matters: all models perform worst on open-space zones and best on mixed/residential contexts.
- **Generalisation to unseen cities is strong** (≤ 3 pp drop), indicating that these tasks probe general visual reasoning rather than city-specific knowledge.

Future directions. Re-running with the fixed pipeline will provide clean accuracy, F1 and AUC numbers for mismatch tasks. Beyond that, the benchmark points to four open research directions: geometry-aware cross-view alignment, spatially grounded semantic counting, direction-aware scene representation, and camera-angle-robust feature extraction — none of which are adequately handled by current pretraining objectives.